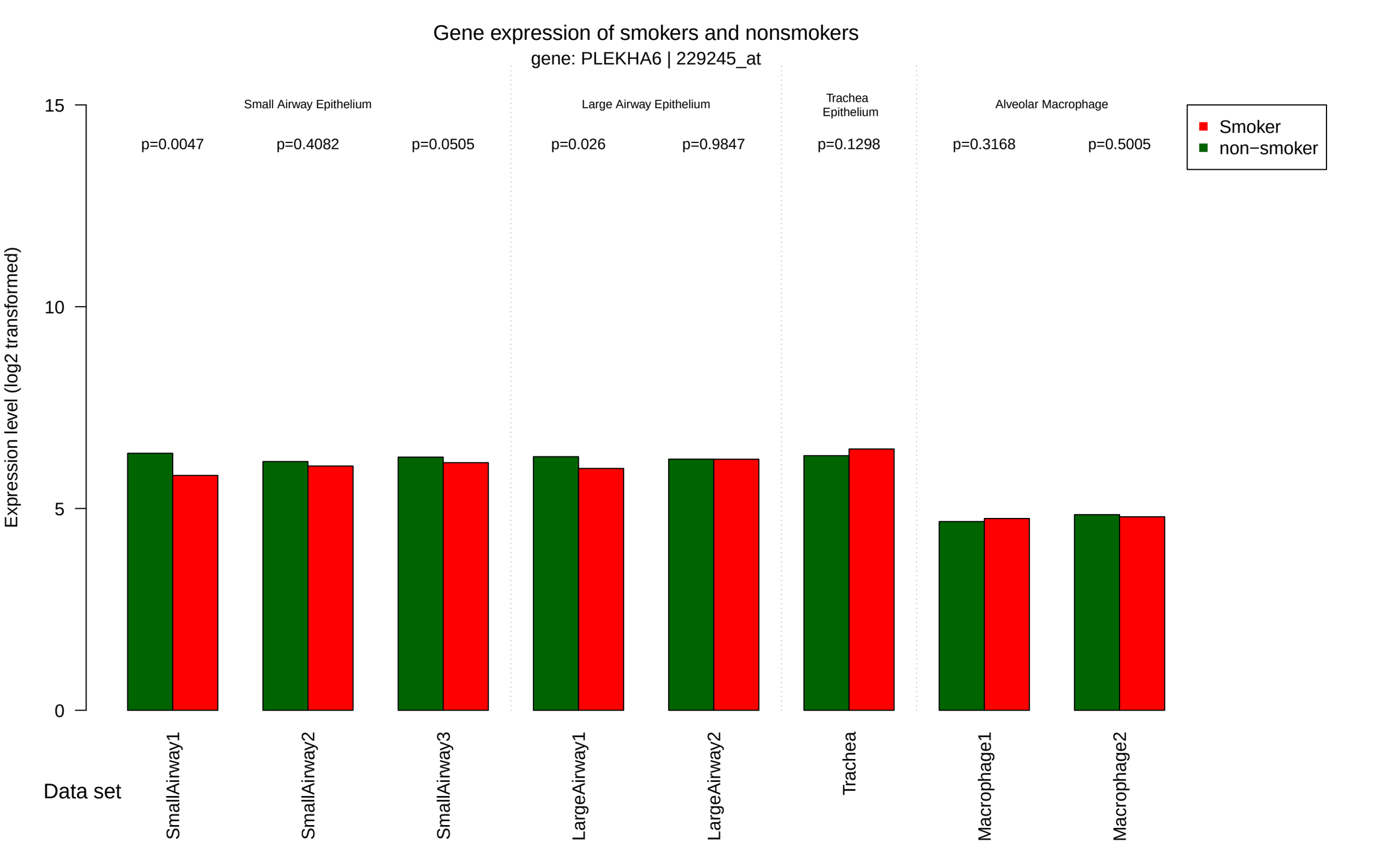
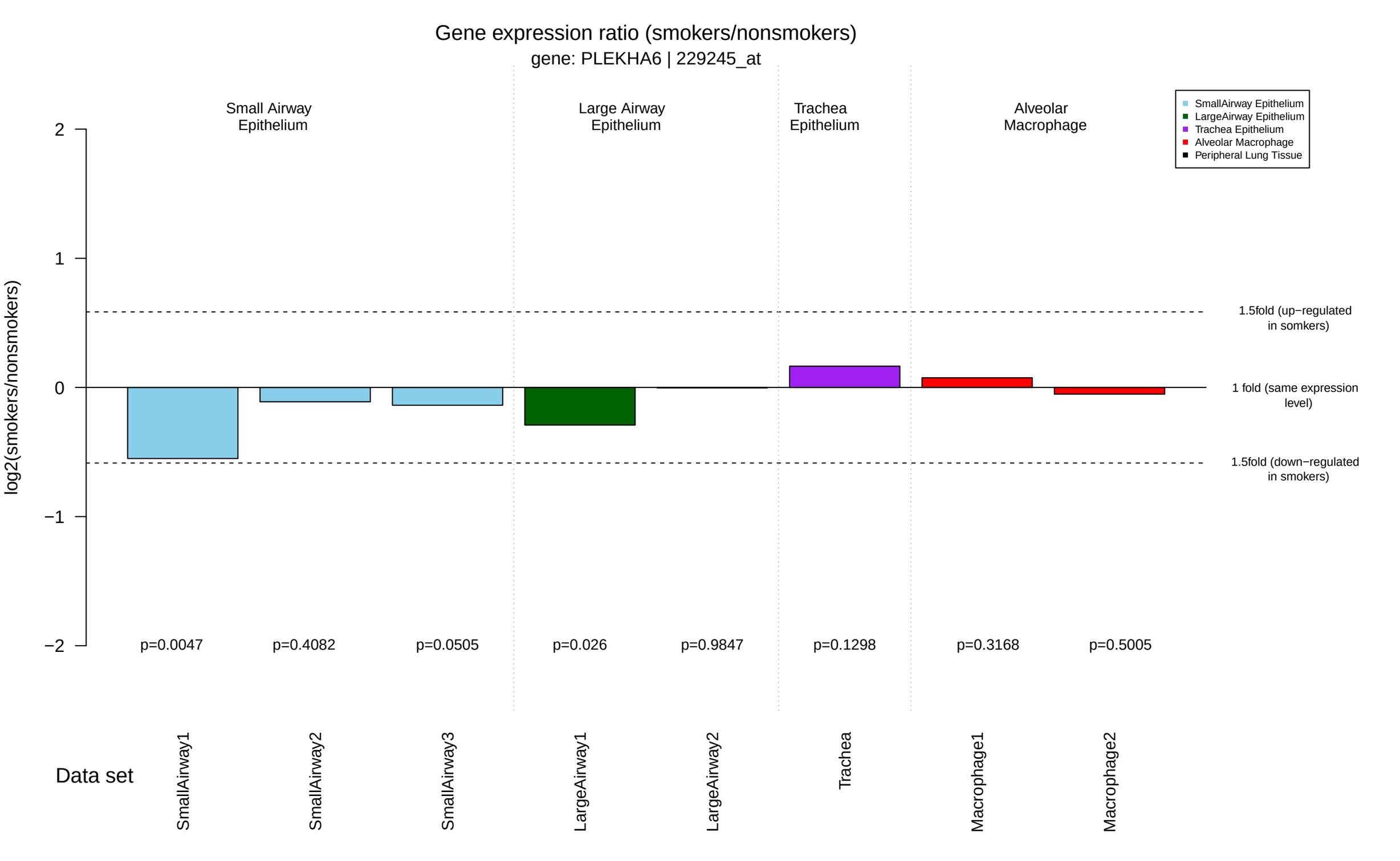
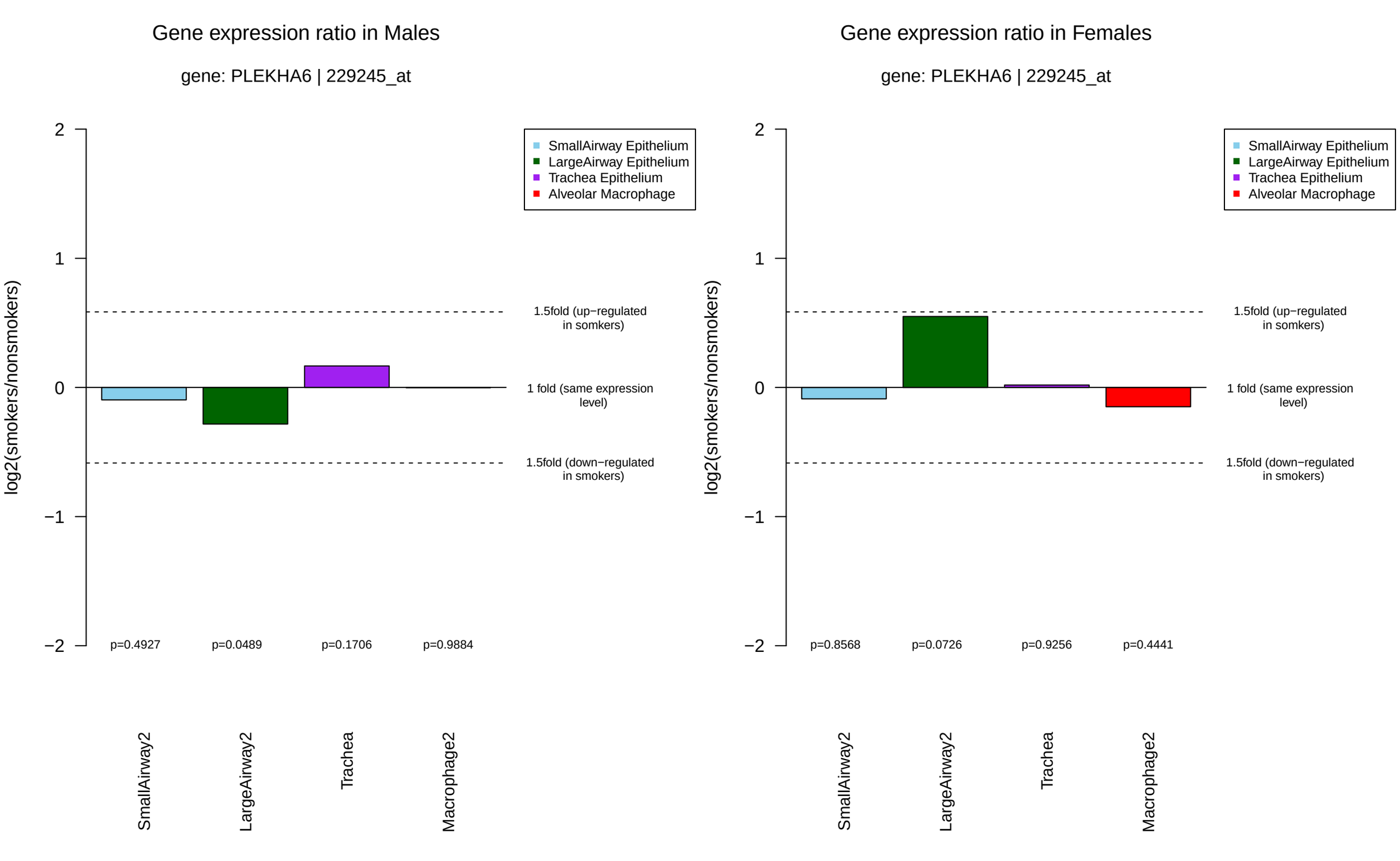
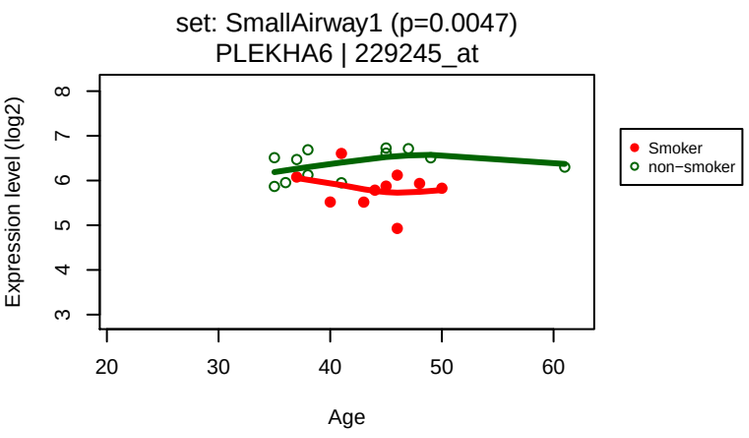
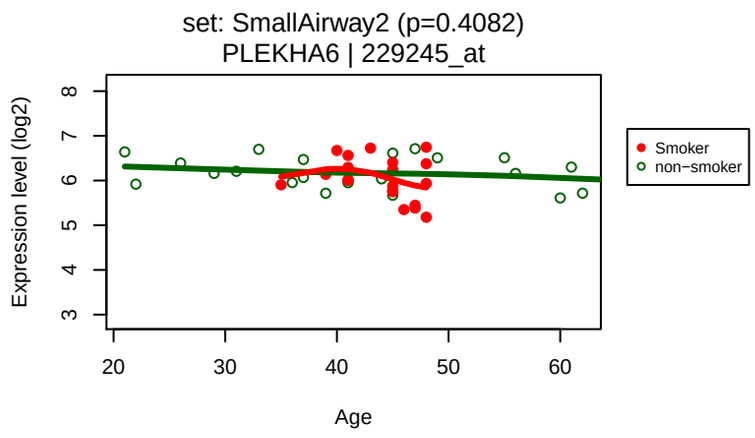
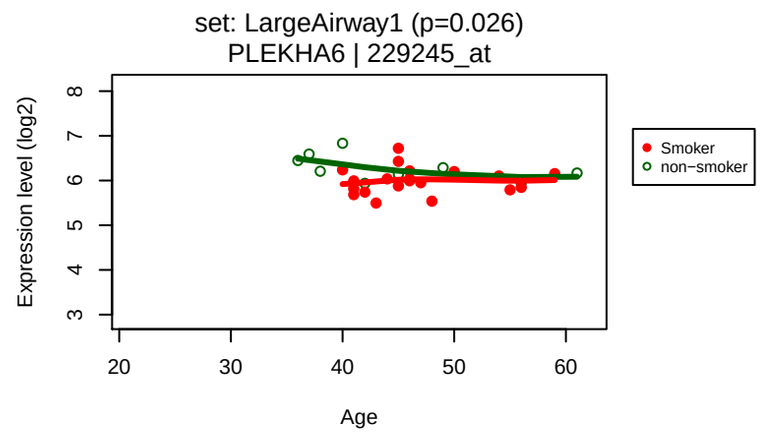

Gene expression of smokers and nonsmokers
gene: PLEKHA6 | 229245_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0047
p=0.4082
p=0.0505
p=0.026
p=0.9847
p=0.1298
p=0.3168
p=0.5005
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

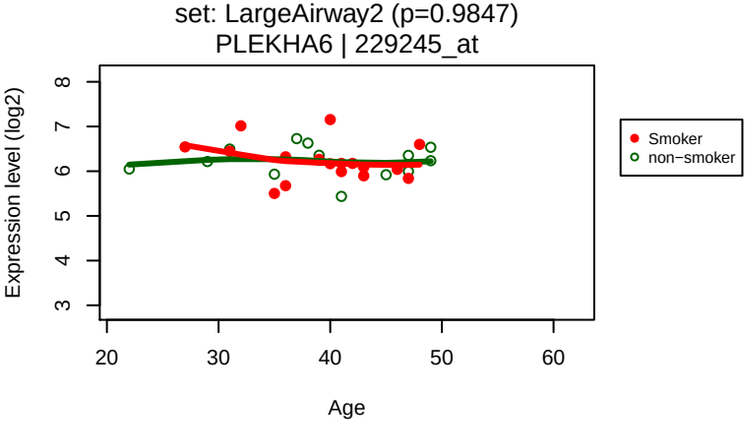

Gene expression ratio (smokers/nonsmokers)
gene: PLEKHA6 | 229245_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0047
p=0.4082
p=0.0505
p=0.026
p=0.9847
p=0.1298
p=0.3168
p=0.5005
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

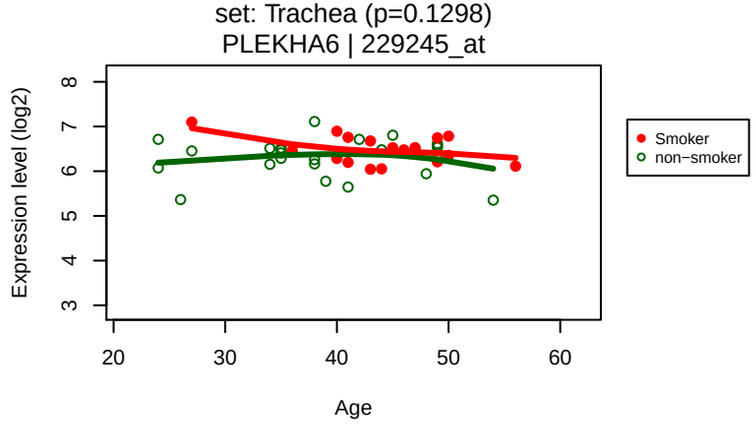

Gene expression ratio in Males
gene: PLEKHA6 | 229245_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4927
p=0.0489
p=0.1706
p=0.9884
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PLEKHA6 | 229245_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8568
p=0.0726
p=0.9256
p=0.4441
SmallAirway2
LargeAirway2
Trachea
Macrophage2

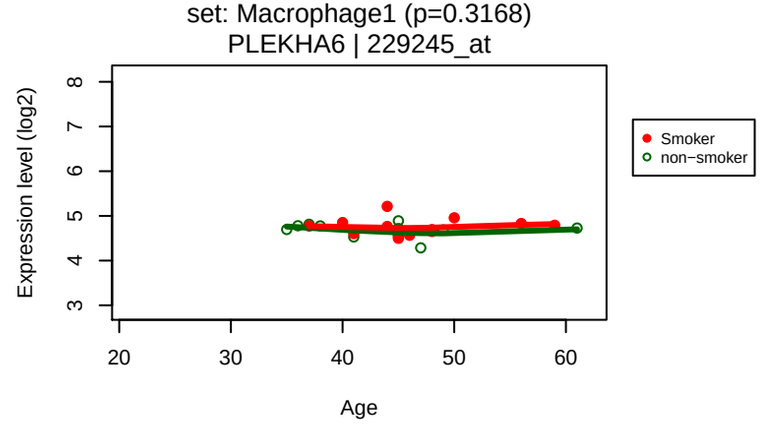

set: SmallAirway1 (p=0.0047)
PLEKHA6 | 229245_at
Expression level (log2)
Age
Smoker
non−smoker

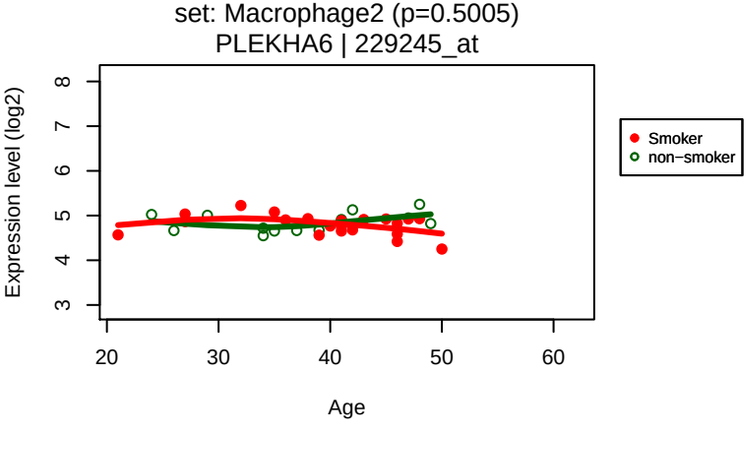

set: SmallAirway2 (p=0.4082)
PLEKHA6 | 229245_at
Expression level (log2)
Age
Smoker
non−smoker

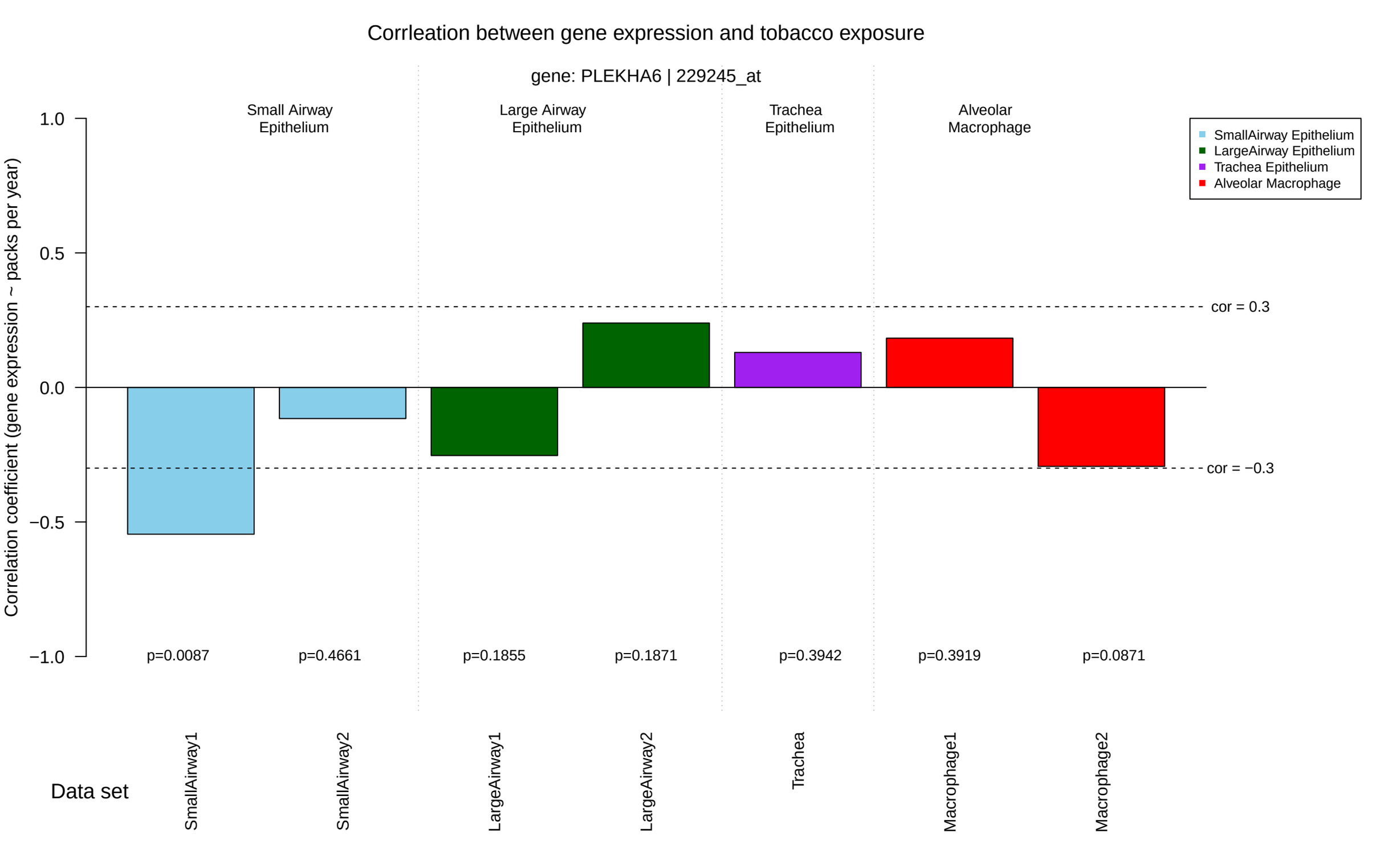

set: LargeAirway1 (p=0.026)
PLEKHA6 | 229245_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9847)
PLEKHA6 | 229245_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1298)
PLEKHA6 | 229245_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3168)
PLEKHA6 | 229245_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5005)
PLEKHA6 | 229245_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PLEKHA6 | 229245_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0087
p=0.4661
p=0.1855
p=0.1871
p=0.3942
p=0.3919
p=0.0871
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2